DM - CLASS TEST KEY 2025

1. **ARM**
   - **Min support count calculation – 1m**
   - **1-Itemsets and Support – 1m**
   - **2-Itemsets and Support – 1m**
   - **3-Itemsets and Support– 2m**
   - **Downward Closure Property – 1m**
   - **Association Rules (8) – 4m**

Support(X) = (Number of transactions containing XXX) / (Total transactions)

1-Frequent Itemset	Support_count
Beach	4
Holiday	2
Ocean	2
Sunshine	4

2-Frequent Itemset	Support_count
Beach, Ocean	2
Beach, Sunshine	3
Holiday, Sunshine	2
Ocean, Sunshine	2

3-Frequent Itemset	Support_count
Beach, Ocean, Sunshine	2

min_support = 40%, ✓

min_support_count

= min_support x itemset_count

= 40% x 5

= 2

We have 5 frequent itemsets:

{Beach, Ocean}, {Beach, Sunshine}, {Holiday, Sunshine}, {Ocean, Sunshine} and {Beach, Ocean, Sunshine}.

Therefore, candidate rules are:

For {Beach, Ocean},

- Beach > Ocean = 2/4 = 50%
- Ocean > Beach = 2/2 = 100% (Strong) ✓

For {Beach, Sunshine},

- Beach > Sunshine = 3/4 = 75% (Strong)
- Sunshine > Beach = 3/4 = 75% (Strong)

For {Holiday, Sunshine},

- Holiday > Sunshine = 2/2 = 100% (Strong) ✓
- Sunshine > Holiday = 2/4 = 50% ✓

For {Ocean, Sunshine},

- Ocean > Sunshine = 2/2 = 100% (Strong) ✓
- Sunshine > Ocean = 2/4 = 50%

For {Beach, Ocean, Sunshine},

- Beach > Ocean, Sunshine = 2/4 = 50%
- Ocean, Sunshine > Beach = 2/2 = 100% (Strong)
- Ocean > Beach, Sunshine = 2/2 = 100% (Strong)
- Beach, Sunshine > Ocean = 2/3 = ~67%
- Sunshine > Beach, Ocean = 2/4 = 50%
- Beach, Ocean > Sunshine = 2/2 = 100% (Strong)

2. FP Growth

- **Item Frequency Table Computation – 1m**
- **Frequent Pattern Set (Support Count) – 1m**
- **Tree Construction – 2m**
- **Conditional Pattern Base, Conditional FP Tree, Frequent Patterns (6 items) – 6m**

Item	Frequency	Item	Frequency
a	3	j	1
b	3	k	1
c	4	l	2
d	1	m	3
e	1	n	1
f	4	o	2
g	1	p	3
h	1	s	1
i	1		

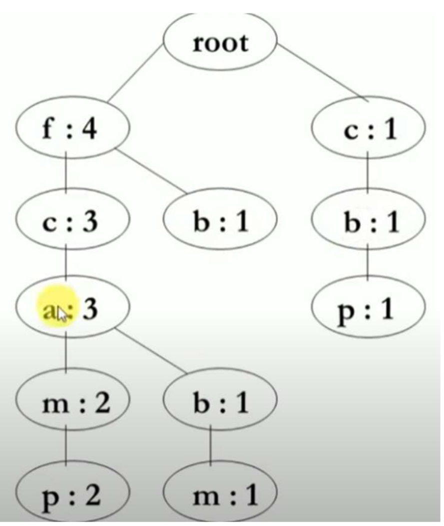
A **Frequent Pattern set (L)** is built, which will contain all the elements whose frequency is greater than or equal to the minimum support. These elements are stored in descending order of their respective frequencies.

As minimum support is 3.

After insertion of the relevant items, the set L looks like this:-

L = { (f:4), (c:4), (a:3), (b:3), (m:3), (p:3) }

Item	Conditional Pattern Base
p	{{f, c, a, m : 2}, {c, b : 1}}
m	{{f, c, a : 2}, {f, c, a, b : 1}}
b	{{f, c, a : 1}, {f : 1}, {c : 1}}
a	{{f, c : 3}}
c	{{f : 3}}
f	Φ



Item	Conditional Pattern Base	Conditional FP-Tree	Frequent Patterns Generated
p	{{f, c, a, m : 2}, {c, b : 1}}	{c : 3}	{<c, p : 3>}
m	{{f, c, a : 2}, {f, c, a, b : 1}}	{f, c, a :3}	{ <f, m : 3>, <c, m : 3> <a, m : 3>, <f, c, m : 3> <f, a, m : 3>, <c, a, m :3>}
b	{{f, c, a : 1}, {f : 1}, {c : 1}}	Φ	{}
a	{{f, c : 3}}	{f, c : 3}	{<f, a : 3>, <c, a : 3>, <f, c, a:3>}
c	{{f : 3}}	{f : 3}	{ <f, c : 3>}
f	Φ	Φ	{}

3. Kmeans

Table – 6m

Euclidean Distance – 1m

New Centroid Calculation - 1.5m + 1.5m = 3m

(6.75 6.5) (2.   3.25)